

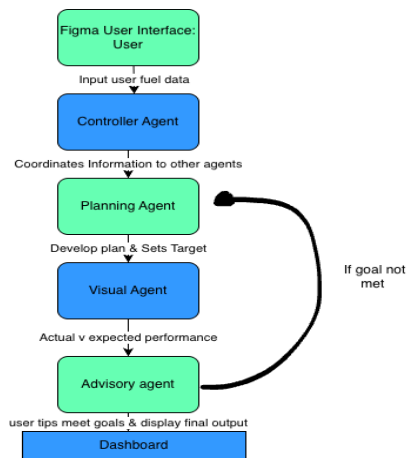
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Agentic AI

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Milestone II: Multi Agent Workflow



The overall purpose of FuelComute is to solve a problem that commuters face frequently.

The users input their usage and can tailor the percentage they wish to see decrease. Following this information source are multiple agents that assist in the journey of generating recommendations, visual tools, and advice. Ensuring the users' needs are met by promoting sustainability, decision-making, and saving.

Agent Roles & Format

This application utilizes multiple agents in helping reach the users goal. The key coordinator is the controller agent in this case serving almost like a manager navigating the workflow of the other agents. The Controller agent feeds the users input and is able to distribute it across the

workflow. The user's input includes the amount of gas used a week(fuel log), and the target goal reduction(percentage). The planning agent is then able to take the user's inputs and develop an actionable plan for the user to achieve their goals. From there the information is passed to the visualization agent that is able to show the amount of fuel increased/decreased from the previous month and another that showcases the current progress of the goal. Lastly, the advisory agent which the UI tailors these agent solutions, and will recommend taking a different form of transportation or leaving early to avoid idling.

Agent Communication

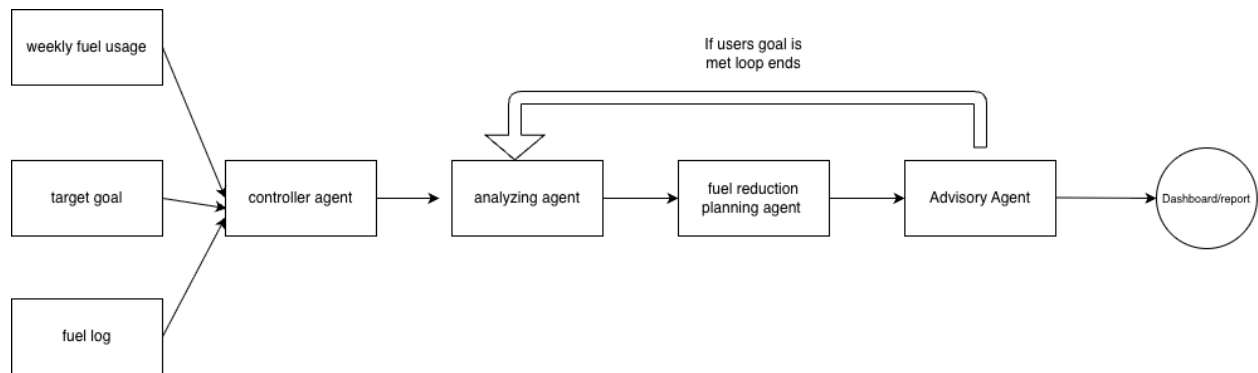
The workflow is a fixed structure, meaning the controller agent coordinates it. This process is sequential because it relies on the previous agent to complete the next action. The first agent that starts the flow, the planning agent, takes the user's input (goal and fuel usage) and generates a plan. Then the visual agent takes that data and plans to visualize a comparison and a goal tracker. The advisory agent takes all the previous agents to evaluate and provide improvements. If the user's goal is not met, the flow will go from the advisory agent to the planning agent to adjust. This will include indicators behind, on track, and goal achieved. From there, it will project what's needed to catch up and provide more actionable tips. The conditions for the loop to stop are that the user's goal has been met or the user modifies inputs.

Check and Adjust

The results are just as good as the information provided. The user should provide accurate information to get reliable and accurate results. The User Interface does have some precautions by not allowing input of negative numbers and also adding a drop-down for the percentage to ensure the goal is obtainable. The application is user-friendly in the sense of transparency; the

user is able to visually see the process that agents take to conclude, and is able to choose what they want to implement.

Multi agent workflow



Mitigation plan

Risks	Design & development of agent	Agent integration into processes
Adversarial attacks	The agent may have similar designs to the other sustainability applications.	This can be integrated by implementing different features to the overall model
Bias and discrimination	The agent can lead to discrimination because of generalized recommendation	Provide AI bias training that updates frequently.
Misinformation and hallucinations	The fuel commute is tailored to the user, but broadly.	This can be enhanced by geolocation as well as realtime tracking
Data privacy violations	The user enters the gas stations they frequent without any protection that their information is protected.	This can be improved by implementing information protection measures

Intellectual property risks	As mentioned before in the data privacy section there are not any safety promotions in the systems design.	Implement those into the agentic system. Utilizing legal resources.
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